



Reading Assignment 1

Report

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Bitcoin: A Peer-to-Peer Electronic Cash System

1 Initial Challenges

- Signature chain vs double-spend prevention. Signatures prove authorization, but I was not clear how the network proves one spend is the first valid one.
- "Longest chain" meaning. I initially treated it as "most blocks" and missed that the core concept is cumulative proof-of-work.
- SPV reliability boundaries. I did not know when Merkle-proof verification is enough and when full-node validation is necessary.
- Section 11 math. I could follow symbols (p, q, z) but not the practical interpretation for confirmation policy.

2 Process of Inquiry

Interaction 1: Why signatures are insufficient without consensus ordering

My Prompt: "In Section 2, a coin is a chain of digital signatures. Why is that still insufficient to prevent double-spending? Give a concrete example." "Show how the timestamp server and proof-of-work combine to solve this gap without a mint."

My Analysis: This response was helpful because it separated two guarantees I had conflated: (1) Signature chain guarantee: only someone with the private key can authorize transfer. (2) Consensus ordering guarantee: the network agrees which conflicting spend came first. The AI used a concrete double-spend example (same UTXO sent to Bob and Carol) to show why valid signatures alone cannot resolve conflict. The key insight was that timestamping + proof-of-work creates a public ordering rule that replaces a mint. I spotted one simplification: it first said "more blocks wins." After follow-up, it corrected to "greatest accumulated proof-of-work wins." That correction was important for technical accuracy.

Interaction 2: SPV security boundaries and realistic trust assumptions

My Prompt: "For Section 8, explain SPV in operational terms: what data does an SPV client store, and what attacks remain possible?" "Under what conditions does SPV remain reliable, and when should a business run a full node instead?"

My Analysis: This exchange turned a short section into a usable threat model. The AI correctly explained that SPV stores block headers and Merkle branches, but does not independently validate full transaction history or all consensus rules. The most useful part was the operational distinction: (1) Occasional receivers may accept SPV with confirmation waiting and network diversity. (2) High-value or frequent-payment merchants should run full nodes for independent rule enforcement and faster fraud detection. I also asked about isolation/eclipsing risks. The AI clearly stated these are modern operational concerns beyond what the whitepaper formalizes, which I considered an honest boundary of the answer.

Interaction 3: Interpreting Section 11 probability and confirmation depth

My Prompt: "Walk through Section 11 calculations conceptually. What do p, q, and z mean in plain language?" "Use the table values to explain whether 6 confirmations are always enough."

My Analysis: This interaction improved my mathematical reading of the paper. The AI mapped variables clearly:- p: probability honest miners find next block. - q: probability attacker finds next block. - z: confirmation depth (how many blocks are added after the payment block). The key insight is that safety is probabilistic, not absolute. Larger z lowers risk exponentially when $p > q$, but required z depends strongly on q. I challenged the common "6 confirmations is always enough" claim. The follow-up correctly used the paper's own table for $P < 0.001$: - q = 0.10 $\rightarrow$ z = 5 ; - q = 0.30 $\rightarrow$ z = 24 ; - q = 0.40 $\rightarrow$ z = 89

3 Synthesis and Lingering Questions

- Bitcoin separates transaction authorization from transaction ordering. Signatures prove spending authority, while proof-of-work consensus determines canonical order and prevents double-spend acceptance.
- Consensus is achieved through a public chain of hashed blocks where rewriting history requires redoing computational work and catching up against ongoing honest mining.
- Security is probabilistic. Confirmations are a risk-management mechanism whose strength depends on attacker hash fraction and waiting depth.
- Incentives (block rewards plus fees) are not an accessory; they are part of the security architecture, aligning miner behavior with chain validity under normal economic conditions